
The HyX-learner: Heterogeneous Treatment Effect Estimation under Sparse Outcomes and Skewed Treatment Distributions

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Abstract

Estimating heterogeneous treatment effects of retail discounts is challenging because sales data are often zero-inflated and discount depths are highly imbalanced. Standard stratified meta-learners suffer from high variance due to these complications. This paper studies conditional average treatment effect estimation and proposes the Hybrid X-learner (HyX-learner), an extension of the Generalised X-learner that combines treatment-specific stratified models with a pooled model through shrinkage based on the size of the treatment samples. To better match the distributional features of retail demand, we incorporate a Tweedie loss for zero-inflated outcomes. Furthermore, we reverse the sign of the regression adjustment in the Generalised X-learner, introducing a Correlated Residual Correction (CRC). We evaluate the framework in a Monte Carlo simulation study with multiple discount levels, and in a real-life application with multiple discount types. Across both standard and excess zero-inflated data generating processes, the HyX-learner consistently outperforms the Generalised X-learner. The strongest gains occur at deep discount levels where treatment sparsity is most severe. The results also show that the Tweedie objective can improve performance, but only when combined with our proposed CRC regression adjustment. In the empirical application, the corrected framework yields economically informative treatment effect rankings evaluated on the GATES criterion. These findings suggest that the HyX-learner provides a computationally feasible and practically useful approach for heterogeneous treatment effect estimation in retail settings with sparse outcomes and skewed treatment distributions.

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1 Introduction

Discounts are paramount in retail pricing strategies as they are frequently applied to stimulate purchase behaviour and manage waste in inventory. Since discounts affect both sales and profits, determining the appropriate depth and timing is of high importance for pricing managers. Evaluating these decisions is essentially a causal inference problem, considering retailers are not solely interested in forecasting sales, but also in understanding how alternative discount levels would have changed sales. Because the effects of these discounts can vary substantially across products and time, capturing this heterogeneity by estimating conditional average treatment effects (CATEs) is necessary for optimal discounting.

However, when estimating causal effects in retail, several challenges arise. First, we do not have access to true counterfactual outcomes, being what would have happened under a different discount. Second, retail data frequently exhibit zero sales, resulting in zero-inflated outcome distributions. Third, discount levels tend to be unevenly distributed across products in time, as pricing managers rely on a limited set of commonly used discount depths, resulting in treatment groups with little observations. Consequently, estimation methods that rely on outcome models stratified by discount depth suffer from high variance when applied to these under-represented treatment groups.

Addressing these challenges requires suited methodological decisions. Traditional approaches relying on parametric models often struggle to capture non-linear heterogeneity that characterizes retail demand. As a result, recent research has increasingly turned to causal machine learning methods for estimating CATEs (Wager & Athey 2018). A central development in this literature is the Double Machine Learning (DML) framework by Chernozhukov et al. (2018a). It combines machine learning with orthogonalization principles, achieving desirable statistical properties such as $\sqrt{n}$ -consistency while controlling for high-dimensional confounding. That is, many covariates that influence both treatment assignment and the outcome of interest. However, DML usually relies on partially linear specifications for treatment effects, an assumption we deem too restrictive in discounting. Furthermore, it relies on propensity score estimation, which may become unstable when treatment probabilities approach zero due to deterministic business rules.

Alternatively, Künzel et al. (2019) propose models that estimate treatment effects by decomposing the causal inference problem into multiple supervised learning tasks, referred to as meta-learners. The most promising variant of these models is the X-learner. Unlike DML, the X-learner does not impose a linear relationship between the treatment variable and its respective causal effect. Because the X-learner relies on outcome modelling instead of propensity weighting, it remains stable even as treatment probabilities approach zero. Furthermore, when treatment distribution is imbalanced, common in retail where shallow discounts occur more often, the X-learner empirically retains high performance (Künzel et al. 2019). While proposed for binary treatments, Acharki et al. (2023) extend the X-learner to account for multiple treatment levels, providing a baseline for estimating heterogeneous treatment effects of discrete discount depths. In addition, they propose a Regression Adjustment (RA) formula for counterfactual prediction that leverages information from the realized treatment level, improving the efficiency of CATE estimation in sparse treatment groups.

While this Generalised X-learner provides a suitable foundation, it remains vulnerable to the statistical properties of retail data. This is because it uses outcome models stratified by treatment level, which can suffer from high estimation variance in under-represented treatment groups. Moreover, minimising regression models on mean squared error (MSE) performs poorly on zero-inflated sales data, as the excess point mass at zero inflates out-of-sample variance. While hurdle models (Mullahy 1986) attempt to solve this by treating zero and positive outcomes as separate processes, the Tweedie distribution provides a unified alternative. It handles both the point mass at zero and the continuous distribution of positive values within a single likelihood framework.

While meta-learning, Tweedie-based estimation, and RA have been studied in isolation, to the best of our knowledge, they have not yet been combined. To address this, we introduce the **Hybrid X-learner** (HyX-learner), which adapts the Generalised X-learner to capture the statistical challenges in retail more effectively. We make three core contributions. First, to address imbalanced treatment groups, we introduce a treatment group-dependent pooling mechanism when estimating the outcome of interest. Motivated by the simultaneous estimation literature of James et al. (1961), we combine the high-variance stratified models with a pooled estimator that includes the treatment variable in the covariate space. This reduces estimation variance for rare discount levels while we theoretically demonstrate that as sample size increases, the HyX-learner inherits all asymptotic guarantees of the Generalised X-learner proven by Acharki et al. (2023). Second, we integrate a Tweedie loss function to model zero-inflated outcomes in a unified framework. Third, we propose a correction to the RA formula of Acharki et al. (2023), called the Correlated Residual Correction (CRC), aligning it with economic intuition.

We evaluate the proposed framework through a simulation study designed to replicate the challenging properties of retail data. The simulations allow us to compare the performance of the proposed HyX-learner with the original Generalised X-learner under different data-generating processes, with known counterfactual outcomes. Our primary finding is that the HyX-learner consistently outperforms the Generalised X-learner in out-of-sample CATE estimation accuracy, regardless of the underlying data-generating process, loss function or RA applied. This underscores the variance-reducing mechanism of our hybrid outcome estimation. Likewise, the CRC outperforms the Acharki et al. (2023) RA formulation across all configurations. Lastly, our findings show a synergy between loss function and RA formula. Specifically, a Tweedie loss combined with the Acharki et al. (2023) RA formulation worsens out-of-sample RMSE relative to using MSE. The Tweedie specification only improves performance relative to MSE when combined with CRC.

Besides a simulation study, we apply the method to a real-world retail dataset to assess its empirical performance. Because true counterfactual outcomes are unknown in this context, we utilise the Grouped Average Treatment Effects (GATES) framework by Chernozhukov et al. (2018b) to assess whether the ranking of the estimated CATEs are consistent with realised treatment effect heterogeneity. The empirical results demonstrate that CRC is essential for correctly ranking observations by their treatment responsiveness, with both meta-learners producing inverted ranking under the Acharki et al. (2023) formula in at least one case.

The remainder of this paper is structured as follows. Section 2 formalises the problem and introduces notation along with necessary assumptions. Section 3 details the methodological framework of the proposed HyX-learner. Section 4 describes the simulation design and empirical evaluation used to assess the performance of the proposed approach. Then, Section 5 discusses the results of the simulation study and the empirical findings. Lastly, Section 6 concludes our work and proposes avenues for further research.

2 Problem Setting

We adopt the Rubin Neyman model Rubin (2005), in which causal effects are defined as comparisons between potential outcomes under different treatment levels. We assume the existence of counterfactuals $Y(d_k)$ as the outcome that would have been observed under treatment $d_k \in \mathcal{D} = \{0, d_1, \dots, d_K\} \subset [0, 1)$, where $\mathcal{D}$ represents $K + 1$ discrete discount levels. Let $\mathcal{I}_j$ denote the set of units i in cluster j ; in a retail context, units are products and clusters are store & product category combinations. Furthermore, let $\mathcal{J}$ and $\mathcal{T}$ be the set of clusters and time periods, respectively. For every $j \in \mathcal{J}$, $i \in \mathcal{I}_j$, and $t \in \mathcal{T}$, we define:

$$\begin{aligned} X_{ijt} &\in \mathbb{R}^p, && \text{the observed characteristics,} \\ D_{ijt} &\in \mathcal{D}, && \text{the observed treatment,} \\ Y_{ijt}(d_k) &\in \mathbb{N}_0, && \text{the potential sales under treatment } d_k \in \mathcal{D}. \end{aligned}$$

Our estimand of interest is the conditional average treatment effect (CATE) of a discount $d_k \in \mathcal{D}$ relative to no discount ($d = 0$), on the outcome of interest:

$$\tau(d_k, X_{ijt}) = \mathbb{E}[Y_{ijt}(d_k) - Y_{ijt}(0) \mid X_{ijt}], \quad \forall d_k \in \mathcal{D}. \quad (1)$$

The CATE can then be used to determine the optimal treatment allocation by e.g. maximizing expected total revenue subject to certain context-specific constraints. To ensure $\tau(d_k, X_{ijt})$ is statistically identified from observational data rather than experimental data, we rely on three identifying assumptions.

Assumption 1, Unconfoundedness. Conditional on X_{ijt} , treatment assignment is independent of the potential outcomes, implying historical discount levels are determined solely by observable characteristics:

$$Y_{ijt}(d_k) \perp D_{ijt} \mid X_{ijt}, \quad \forall d_k \in \mathcal{D}.$$

In retail, this assumption is challenged by unobserved managerial anticipation of demand shocks, such as timing promotions for seasonal events or reacting to unsatisfactory past sales. If the drivers of these demand shocks remain unobserved, estimated treatment effects will be biased. To mitigate this confounding, we condition on a rich covariate space X_{ijt} which is also the standard in the literature.

Assumption 2, Overlap. Every unit of observation has a non-zero probability of receiving any treatment level, ensuring that for any given profile of observed characteristics X_{ijt} , we can estimate treatment effects without having to extrapolate to unseen regions of the data:

$$\mathbb{P}(D_{ijt} = d_k \mid X_{ijt}) > 0, \quad \forall d_k \in \mathcal{D}.$$

This assumption is challenged by deterministic pricing rules. Retailers may designate low margin products to never be subjected to deep discounts or reserve deep discounts exclusively for perishable products, driving treatment propensity scores near zero for some units. While such overlap violations severely destabilise traditional estimation methods that rely on inverse probability weighting estimators, our method mitigates this vulnerability.

Assumption 3, Stable Unit Treatment Value Assumption (SUTVA). The potential outcome of each observation depends only on its own treatment assignment and is unaffected by the treatment assignments of other observations:

$$Y_{ijt}(d_k) = Y_{ijt}(d_k, \mathbf{D}_{-ijt}), \quad \forall \mathbf{D}_{-ijt}, \quad \forall d_k \in \mathcal{D}, \quad (2)$$

where $\mathbf{D}_{-ijt}$ denotes the vector of treatment assignment of all units other than (i, j, t) . In retail, concepts such as substitution effects and stockpiling, referred to as spatio-temporal interference, can violate this assumption. For instance, discounting a specific winter jacket typically cannibalizes sales of substitute jackets. To address this, we evaluate the robustness of our proposed estimators against violations with a stress test, detailed in Section 4.1.2.

3 Methods

This section describes the methodological components of the proposed HyX-learner (**Hy**brid **X**-learner). We begin by introducing the Generalised X-learner of Acharki et al. (2023) in Section 3.1, the foundational framework upon which all subsequent contributions build. Section 3.2 then proposes the Hybrid X-Learner, which modifies the first stage of this framework to address the high estimation variance that arises in sparse treatment groups. Section 3.3 introduces a second first-stage adaptation: replacing the standard MSE loss with a Tweedie loss to handle the zero-inflated structure of retail sales data. Section 3.4 proposes a correction to the second-stage crossover imputation, aligning the regression adjustment term with economic intuition. Finally, Section 3.5 motivates the choice of XGBoost as the supervised learner underlying all regression tasks in the framework.

3.1 The Generalised X-learner

The Generalised X-learner, specifically the Regression-Adjustment formulation of Acharki et al. (2023), extends the binary X-learner to a multi-valued treatment setting while avoiding the insta-

bility associated with propensity score weighting in their naïve extension of the binary case. The framework estimates CATEs by separating outcome estimation from treatment-effect estimation through a crossover imputation strategy, justifying the ‘X’ in X-learner. The estimation procedure consists of three stages.

Stage 1: Outcome Estimation. We stratify the data into subsamples S_k corresponding to each treatment level d_k , such that $S_k = \{(i, j, t) : D_{ijt} = d_k\}$ for all $k \in \{0, \dots, K\}$. For each treatment level we train a separate machine learning (ML) model $\hat{\mu}_{d_k}(x)$, trained exclusively on S_k , estimating the conditional mean of the outcome:

$$\hat{\mu}_{d_k}(x) = \mathbb{E}[Y_{ijt}(d_k) \mid X_{ijt} = x]. \quad (3)$$

Stratifying the models in this way is essential to avoid regularisation bias in the estimated treatment effects (Chernozhukov et al. 2018a). A single model trained on all treatment levels simultaneously, known as an S-learner (Künzel et al. 2019), appends the treatment variable to the covariates. Because the regulariser cannot distinguish the treatment dimension from any other covariate, the treatment effect is implicitly downweighted, inducing a systematic bias toward zero that does not vanish unless the treatment signal is large relative to competing confounders.

In parametric regularised models like Ridge regression, this bias can be avoided by excluding the treatment coefficient from the penalty term. However, this is not possible in tree-based methods like XGBoost. Regularisation in gradient boosted trees operates through structural constraints (maximum tree depth, minimum child weight, learning rate) that determine overall complexity rather than penalising individual features, with no method to exempt specific ones. Therefore, the treatment variable tends to be under-represented, as its effect is modest compared to the variance explained by unit-level covariates in retail data (e.g., price likely has a comparable marginal variance reduction to discount). By fitting a dedicated model on each stratum S_k , stratification removes the treatment indicator from the feature space entirely; regularisation can no longer attenuate the treatment contrast, and the estimators in (3) are free of this bias.

Stage 2: Crossover Imputation. Using the Stage 1 models to estimate counterfactual outcomes, we construct a pseudo-outcome Z_{ijt}^k for each target treatment level d_k . Here, the term *pseudo-outcome* refers not to the outcome variable Y_{ijt} itself, but to an imputed treatment effect: a per-observation proxy for $\tau(d_k, x) = \mu_{d_k}(x) - \mu_{d_0}(x)$ that can be used as a dependent variable in the succeeding stage. Acharki et al. (2023) distinguish three cases:

$$Z_{ijt}^k = \begin{cases} Y_{ijt} - \hat{\mu}_{d_0}(x), & \text{if } D_{ijt} = d_k, \\ \hat{\mu}_{d_k}(x) - Y_{ijt}, & \text{if } D_{ijt} = d_0, \\ (\hat{\mu}_{d_k}(x) - \hat{\mu}_{d_0}(x)) + (\hat{\mu}_{d_l}(x) - Y_{ijt}), & \text{if } D_{ijt} = d_l, d_l \in D \setminus \{d_k, d_0\}. \end{cases} \quad (4)$$

Note that when neither the target treatment d_k nor the control d_0 is observed, both counterfactual outcomes are imputed and a regression adjustment is applied. This retains the full sample size when constructing pseudo-outcomes, improving statistical efficiency, and eliminates the need for inverse propensity weighting, making the procedure more robust to violations of the overlap assumption in Section 2. Combining the three cases with indicator functions yields the compact formula:

$$Z_{ijt}^k = \mathbf{1}\{D_{ijt} = d_k\}(Y_{ijt} - \hat{\mu}_{d_0}(x)) + \sum_{l \neq k} \mathbf{1}\{D_{ijt} = d_l\}[(\hat{\mu}_{d_k}(x) - \hat{\mu}_{d_0}(x)) + (\hat{\mu}_{d_l}(x) - Y_{ijt})]. \quad (5)$$

Stage 3: CATE Estimation. We estimate the CATE for each treatment level by regressing the constructed pseudo-outcomes on the covariates X_{ijt} using a separate ML model per treatment level, trained by minimising MSE:

$$\hat{\tau}_{d_k}(x) = \hat{\mathbb{E}}[Z_{ijt}^k \mid X_{ijt} = x], \quad \forall d_k \in \mathcal{D}.$$

The pseudo-outcome construction in Stage 2 ensures that all observations contribute to estimating each $\hat{\tau}_{d_k}(\cdot)$, even when only a minority of units were actually assigned to d_k . This is particularly valuable in retail, where deep discount levels occur infrequently.

3.2 The Hybrid X-learner

The Generalised X-learner described above trains a separate outcome model on each stratum S_k , which eliminates regularisation bias but introduces high estimator variance when these strata are small. This variance propagates through the crossover imputation step in Stage 2, consequently inflating the variance of the pseudo-outcomes in (5). As a result, the Stage 3 CATE estimators $\hat{\tau}_{d_k}(\cdot)$, which are regressed on these values, suffer from decreasing out-of-sample estimation accuracy. This holds particularly for the high-discount treatment groups, which are often of great commercial relevance.

To mitigate this, we hybridise the Stage 1 outcome models, by replacing the stratified learners with a sample-size-adaptive convex combination of two complementary estimators: a stratified model, trained separately per stratum, and a pooled model trained jointly on all observations with the treatment variable as a covariate. The latter yields lower variance predictions by leveraging the full sample, also for under-represented treatment groups. This comes with the cost of a potentially higher bias as outlined in Section 3.1. The convex combination is designed to approximate the optimal trade-off between this bias and variance. Stages 2 and 3 of the estimation procedure remain entirely unchanged, only the Stage 1 outcome models are modified.

The remainder of this subsection proceeds as follows. We first introduce the pooled outcome model formally, and then define the mathematical underpinnings of the hybrid Stage 1 estimator. We conclude with a statistical justification grounded in James-Stein shrinkage theory (James et al. 1961) and an error-propagation argument linking Stage 1 accuracy to Stage 3 CATE quality.

Pooled Outcome Model. Let $N = \sum_{k=0}^K n_k$ denote the total number of training observations, where $n_k = |S_k|$. The pooled model is trained on the full dataset by augmenting the feature space with the treatment variable. Formally, let $\tilde{X}_{ijt} = (X_{ijt}^\top, D_{ijt})^\top \in \mathbb{R}^{p+1}$ be the augmented covariate vector. Then, the pooled model estimates the following conditional mean:

$$\hat{\mu}^{(S)}(x, d_k) = \hat{\mathbb{E}}[Y_{ijt} \mid \tilde{X}_{ijt} = x],$$

where the S in $\hat{\mu}^{(S)}(x, d_k)$ refers to 'Single'.

Hybrid Stage 1 Estimator. Having introduced the single pooled model, we define the hybrid outcome model for treatment level d_k as a convex combination of the stratified model $\hat{\mu}_{d_k}(x)$, as defined in (3), and the pooled model $\hat{\mu}^{(S)}(x, d_k)$:

$$\hat{\mu}_{d_k}^{(H)}(x) = \alpha_k \hat{\mu}_{d_k}(x) + (1 - \alpha_k) \hat{\mu}^{(S)}(x, d_k).$$

Note that we need to estimate or tune the weights α_k . To this extent, we analyse the theoretically optimal expression by inspecting α_k^* which minimises the expected MSE of the hybrid estimator. For now we neglect indices for notational convenience. Let the prediction errors of the outcome models be defined as $e_{d_k}^{(T)}(X) = \hat{\mu}_{d_k}(X) - \mu_{d_k}(X)$ and $e_{d_k}^{(S)}(X) = \hat{\mu}^{(S)}(X, d_k) - \mu_{d_k}(X)$. Let $\sigma_{T,k}^2 = \mathbb{E}_X[e_{d_k}^{(T)}(X)^2]$ and $\sigma_{S,k}^2 = \mathbb{E}_X[e_{d_k}^{(S)}(X)^2]$ denote the marginal expected squared errors of the stratified and pooled models, respectively, integrated over the covariate distribution. Furthermore, let $\sigma_{TS,k} = \mathbb{E}_X[e_T(X)e_S(X)]$. Then, the expected squared error of the hybrid estimator evaluates to:

$$\mathbb{E}_X[(\hat{\mu}_{d_k}^{(H)}(X) - \mu_{d_k}(X))^2] = \alpha_k^2 \sigma_{T,k}^2 + (1 - \alpha_k)^2 \sigma_{S,k}^2 + 2\alpha_k(1 - \alpha_k)\sigma_{TS,k}. \quad (6)$$

Taking the derivative of (6) with respect to α_k and setting it to zero yields the optimal weight:

$$\alpha_k^* = \frac{\sigma_{S,k}^2 - \sigma_{TS,k}}{\sigma_{T,k}^2 + \sigma_{S,k}^2 - 2\sigma_{TS,k}}. \quad (7)$$

An important requirement of the hybrid approach is that it must asymptotically converge to the purely stratified estimator, in order to prevent the regularisation bias in large samples. Using (7), we formalise this property below.

Theorem 1. *Assume that the stratified model $\hat{\mu}_{d_k}(x)$ is a consistent estimator of $\mu_{d_k}(x)$. Furthermore, assume the pooled model maintains a strictly positive, bounded expected squared error asymptotically, such that $0 < \lim_{N \rightarrow \infty} \sigma_{S,k}^2 < \infty$. Then, as $n_k \rightarrow \infty$ such that $N \rightarrow \infty$, the optimal combination weight $\alpha_k^* \rightarrow 1$.*

Proof. By the Cauchy-Schwarz inequality, which states that for any two random variables X and Y , it holds that

$$\mathbb{E}[XY] \leq \sqrt{\mathbb{E}[X^2]}\sqrt{\mathbb{E}[Y^2]} \iff \mathbb{E}[XY]^2 \leq \mathbb{E}[X^2]\mathbb{E}[Y^2].$$

As a result, the squared cross-moment $\sigma_{TS,k}^2$ is strictly bounded by the product of the expected squared errors. That is,

$$\sigma_{TS,k}^2 = \mathbb{E}_X[e_T(X)e_S(X)]^2 \leq \mathbb{E}_X[e_T(X)^2]\mathbb{E}_X[e_S(X)^2] = \sigma_{T,k}^2\sigma_{S,k}^2.$$

As $n_k \rightarrow \infty$, the consistency of the stratified model implies $\sigma_{T,k}^2 \rightarrow 0$. Since $\sigma_{S,k}^2$ converges to a strictly positive constant reflecting the squared asymptotic bias, it follows that:

$$\lim_{n_k \rightarrow \infty} \sigma_{TS,k}^2 \leq 0 \cdot \lim_{N \rightarrow \infty} \sigma_{S,k}^2 = 0 \implies \lim_{n_k \rightarrow \infty} \sigma_{TS,k} = 0.$$

Substituting these limits into the expression for the optimal weight in (7) yields:

$$\lim_{n_k \rightarrow \infty} \alpha_k^* = \frac{\lim_{N \rightarrow \infty} \sigma_{S,k}^2 - 0}{0 + \lim_{N \rightarrow \infty} \sigma_{S,k}^2 - 0} = 1. \quad \square$$

As a result of Theorem 1, the HyX-learner is asymptotically equivalent to the Generalised X-learner of Acharki et al. (2023). Consequently, it inherits all of their asymptotic results and theoretical guarantees without modification. While (7) defines the theoretical optimum, the true expected squared errors $\sigma_{T,k}^2$ and $\sigma_{S,k}^2$ are unknown and difficult to estimate considering we are dealing with small treatment groups. We therefore propose a specification for α_k that mimics the finite-sample behaviour of the optimal weight using mostly the structural parameters of the dataset. Specifically, we use

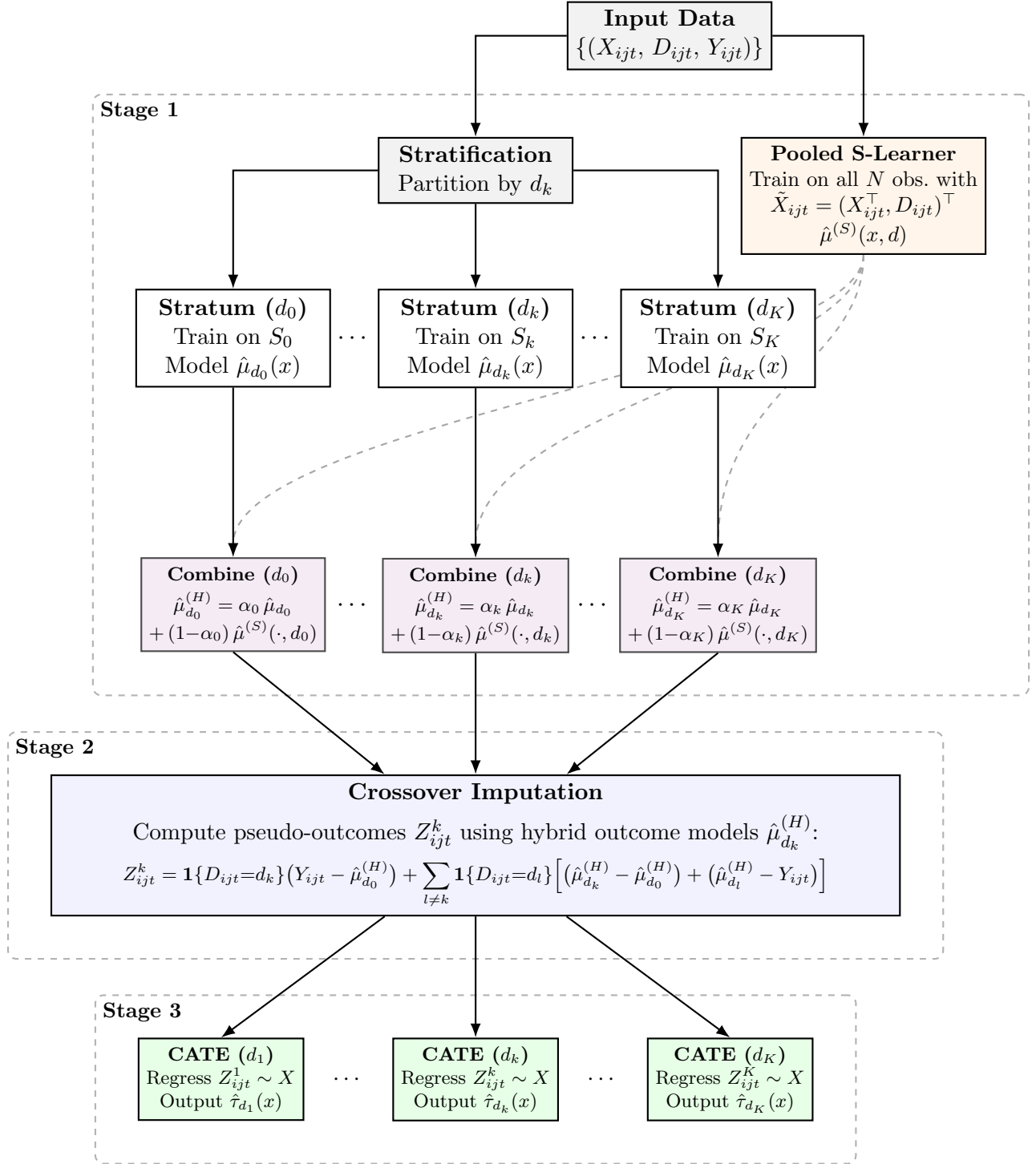
$$\alpha_k = \frac{n_k}{n_k + \delta \sqrt{\frac{N}{K+1}}}, \quad n_k = |S_k|.$$

This functional form preserves several desirable properties. First, α_k is strictly increasing in n_k : treatment levels with more observations automatically receive more weight on the low-bias stratified model. Second, $\alpha_k \rightarrow 0$ as $n_k \rightarrow 0$, so completely unobserved treatment levels default entirely to the pooled model. Third, the denominator's penalty term grows strictly sub-linearly with n_k . Hence, as $n_k \rightarrow \infty$, the linear n_k term dominates, ensuring $\alpha_k \rightarrow 1$ and preserving the theoretical guarantees established in Theorem 1. Note, δ is a hyperparameter which we select using a grid search, for which details are provided in Appendix A.1.

To understand its behaviour concretely, consider a retail setting with $K+1 = 9$ treatment levels and $N = 9,000$ total observations, with $\delta = 10$. The threshold evaluates to $10 \cdot \sqrt{9,000/9} \approx 316$. For an 80% discount group that is applied rarely and contains only $n_k = 50$ observations, the weight is $\alpha_k = 50/(50 + 316) \approx 0.13$. The pooled model acts as a strong stabiliser, receiving 87% of the weight to address the high variance of the small stratum.

The hybrid outcome model $\hat{\mu}_{d_k}^{(H)}(\cdot)$ replaces $\hat{\mu}_{d_k}(\cdot)$ everywhere in (3), (4) and (5). For reference, Figure 1 below provides a visual overview of the full three-stage procedure of the HyX-learner. Note, the Generalised X-learner of Acharki et al. (2023) is recovered as a special case by setting $\alpha_k = 1$ for all treatment levels k , which reduces the hybrid Stage 1 estimator to the purely stratified model.

Figure 1: Flowchart of the Hybrid X-learner. Stage 1 trains $K + 1$ stratified models and one pooled S-learner, then combines them via weights α_k to form the hybrid outcome models $\hat{\mu}_{d_k}^{(H)}(\cdot)$. Stages 2 and 3 are unchanged from the Generalised X-learner.



Statistical Justification. The theoretical motivation for the Hybrid X-Learner rests on a classical result from the simultaneous estimation literature. Specifically, estimating $K + 1$ treatment-level outcome models $\{\mu_{d_k}\}_{k=0}^K$ simultaneously resembles the classical problem analysed by James et al. (1961). Their foundational result establishes that for $K + 1 \geq 3$, any estimator that shrinks group estimates toward a pooled value strictly dominates the unstable per-group estimator in total expected squared loss. This implies the existence of combination weights $\alpha_k^* \in (0, 1)$ such that:

$$\sum_{k=0}^K \mathbb{E} \left[\left\| \hat{\mu}_{d_k}^{(H)} - \mu_{d_k} \right\|^2 \right] < \sum_{k=0}^K \mathbb{E} \left[\left\| \hat{\mu}_{d_k}^{(T)} - \mu_{d_k} \right\|^2 \right]. \quad (8)$$

In our formulation, the pooled model serves as the shrinkage target. Since our simulation employs $K + 1 = 9$ treatment levels, the dominance condition $K + 1 \geq 3$ is satisfied. We recognise that (8) is a statement about total MSE summed over all treatment levels, whereas individual treatment levels with large n_k may experience a negligible increase, but our findings demonstrate this is more than offset by the gains on treatments with small n_k .

Error Propagation to CATE Estimates. The Stage 1 MSE reduction in (8) translates directly to reduced noise in the Stage 2 pseudo-outcomes, and thereby into improved Stage 3 CATE estimates. Consider the pseudo-outcome for an observation assigned to treatment $d_l \notin \{d_0, d_k\}$:

$$Z_{ijt}^k = \left(\hat{\mu}_{d_k}^{(H)}(x_{ijt}) - \hat{\mu}_{d_0}^{(H)}(x_{ijt}) \right) + \left(\hat{\mu}_{d_l}^{(H)}(x_{ijt}) - Y_{ijt} \right).$$

Writing $\hat{\mu}_{d_k}^{(H)} = \mu_{d_k} + \varepsilon_k^{(H)}$, the deviation of Z_{ijt}^k from the true CATE in (1) equals:

$$Z_{ijt}^k - \tau(d_k, x_{ijt}) = \underbrace{\varepsilon_k^{(H)} - \varepsilon_0^{(H)}}_{\text{Stage 1 error}} + \underbrace{\varepsilon_l^{(H)} + \mu_{d_l}(x_{ijt}) - Y_{ijt}}_{\text{irreducible noise}}.$$

Every Stage 1 error term contributes to the training noise of the Stage 3 CATE models. Since the HyX-Learner can reduce $\text{Var}[\varepsilon_k^{(H)}]$ for treatment groups with small sample sizes, the variance of the pseudo-outcome targets is reduced accordingly. As the Stage 3 estimator minimises a squared loss over these targets, lower target variance directly improves out-of-sample CATE squared error. This underscores the theoretical grounding for the empirical squared error improvements we observe in our simulation study.

3.3 Sparse Outcomes

Having described the Hybrid X-Learner’s modification to address sparse treatment groups, we introduce a second Stage 1 adaptation targeting the zero-inflated, right-skewed distribution of retail sales. While minimising the MSE yields asymptotically unbiased estimators of the conditional expectation regardless of the underlying distribution (Hastie et al. 2009), it implicitly assumes homoskedastic errors. In retail sales, conditional variance routinely scales with the conditional

mean. Consequently, using the MSE is statistically inefficient considering it fails to minimise the variance of the outcome model in finite samples, threatening the bias-variance trade-off that defines the quality of the pseudo-outcomes and final CATE estimates. To overcome this inefficiency, we adopt the Tweedie loss, based on the Tweedie subset of exponential dispersion models (Tweedie et al. 1984, Jørgensen 1987). The Tweedie family parameterises heteroskedasticity via

$$\text{Var}(Y) = \phi \mu^p,$$

where $\mu = \mathbb{E}[Y]$, $\phi > 0$ is the dispersion parameter, and p is the power index. We set $p = 1.5$, corresponding to the Compound Poisson-Gamma distribution. As detailed by Jørgensen (1987), this formulation naturally accommodates the data-generating process of retail sales. It captures exact zeros (via a Poisson arrival process) and right-skewed continuous positive outcomes (via Gamma-distributed variables) without requiring a separate hurdle component or individual parameter estimation. Because the dispersion parameter ϕ acts only as a constant scaling factor, we train the outcome models by directly minimising the Tweedie unit deviance for $1 < p < 2$ (Yang et al. 2018):

$$\mathcal{L}_{\text{tw}}(y, \hat{\mu}) = 2 \left(\frac{\max(y, 0)^{2-p}}{(1-p)(2-p)} - \frac{y \hat{\mu}^{1-p}}{1-p} + \frac{\hat{\mu}^{2-p}}{2-p} \right).$$

Minimising this deviance is asymptotically equivalent to a weighted regression that effectively down-weights high-variance observations. This yields a more efficient estimator than the MSE for sparse, zero-inflated data, reducing the noise propagated to the crossover imputation and the final CATE estimates.

3.4 Regression Adjustment

Having adapted Stage 1 of the X-learner through the hybrid outcome models and the Tweedie loss, we now turn to Stage 2 and propose a correction to the crossover imputation formula. Specifically, we revisit the residual-adjustment term in the third case of (4) and argue that its sign, as stated in Acharki et al. (2023), is counter-intuitive from a business economics perspective.

In particular, consider imputing the treatment effect of a 50 % discount (d_k) for an observation where a 25 % discount (d_l) was actually applied. The pseudo-outcome is constructed by imputing both sales under no discount and under 50 % discount and taking their difference, then adding a residual adjustment based on the prediction error at the observed 25 % discount level. In Acharki et al. (2023), this residual is defined as $\hat{\mu}_{d_l}(x) - Y_{ijt}$. When the model overestimates at the observed discount level, the term is positive and the pseudo-outcome is adjusted upward. However, we argue that this operates in the wrong direction. An overestimation at d_l suggests that the estimated treatment effects are systematically biased upward and the pseudo-outcome should be corrected downward. This is because we expect prediction errors to be positively correlated as unobserved factors (marketing campaigns, demand shocks) affect treatment effects similarly across all non-control discount levels. Importantly, as this does not rely on the ordinal nature of discounts but solely on correlated prediction errors, it is applicable to treatments in general.

To this extent, we reverse the sign of the regression-adjustment term, and refer to this form as Correlated Residual Correction (CRC). The CRC pseudo-outcome formula for the case $D_{ijt} = d_l \in D \setminus \{d_0, d_k\}$ therefore becomes:

$$Z_{ijt}^k = (\hat{\mu}_{d_k}(x) - \hat{\mu}_{d_0}(x)) - (\hat{\mu}_{d_l}(x) - Y_{ijt}).$$

3.5 Base Learners

Having described all methodological adaptations of the Generalised X-learner, we conclude this section by specifying the supervised ML model that underlies all regressions in Figure 1. Both for the outcome estimation in Stage 1 and the CATE estimation in Stage 3, we employ Extreme Gradient Boosting (XGBoost) regressors (Chen 2016). This decision is supported by the following two arguments.

First, retail sales data is inherently tabular and typically contains a mixture of nominal variables (e.g. product groups and store clusters), continuous variables (e.g. prices), and ordinal variables (e.g. stock levels). Tree-based methods have been shown to be particularly effective for tabular data (Grinsztajn et al. 2022), as they capture complex feature interactions without the need for advanced embeddings, which deep learning methods generally need to perform competitively.

Second, gradient boosting is uniquely well-suited to accommodate the Tweedie adaptation described in Section 3.3. Because XGBoost sequentially fits decision trees to the negative gradients of an objective function of choice, the outcome models in Stage 1 can be trained directly on the Tweedie unit deviance, inheriting the variance-weighting properties discussed in Section 3.3. Alternative tree-based ensembles that rely on bootstrap aggregation, such as Random Forests, do not admit an arbitrary differentiable loss and therefore cannot incorporate the Tweedie objective in the same manner.

4 Experiments

This Section describes how we evaluate the proposed methods across two experimental designs. First, we benchmark all estimators against known CATEs in a Monte Carlo simulation study, which is outlined in Section 4.1. Second, Section 4.2 describes how we apply our methods to real-life data and assess whether the estimated CATE rankings align with realised treatment effects, without the need for unknown counterfactual outcomes (Chernozhukov et al. 2018b). For hyperparameter selection and simulation details we refer to Appendix A.2 and B, respectively.

4.1 Simulation Study

This Section describes our simulation experiment to assess the contribution of each model component to out-of-sample performance. First, global simulated data characteristics are outlined, after which we provide the mathematical underpinnings of the Data Generating Processes (DGPs) used. Then, the estimators trained and evaluated on these DGPs are defined, and lastly we introduce the metrics to assess their performance.

4.1.1 Global Data Characteristics

The simulated data follow the panel structure formalised in Section 3. We generate $J = 5$ clusters, each containing $n_j = 50$ product units, observed over $T = 50$ time periods, yielding a panel of $N = J \cdot n_j \cdot T = 12,500$ observations. The treatment set consists of $K + 1 = 9$ uniformly-spaced discount levels:

$$\mathcal{D} = \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8\},$$

spanning the full range of practically relevant retail promotions. Each observation is described by covariates $X_{ijt} \in \mathbb{R}^p$, where $p = 20$. Nineteen of these are drawn independently from a standard normal distribution. The final covariate is the natural logarithm of the unit’s price, $\log P_{ij}$, which is time-invariant and is drawn from cluster-specific log-normal distributions.

Data are partitioned using a time-ordered split that preserves the panel structure. The first $T_{\text{train}} = 40$ time periods for all units form the training set ($N_{\text{train}} = 10,000$ observations), and the remaining $T_{\text{test}} = 10$ periods form the test set ($N_{\text{test}} = 2,500$ observations). This temporal ordering is preserved to prevent look-ahead bias. All model fitting and tuning is performed exclusively on the training set, whereas evaluation is performed solely on the test set. The full experimental design is repeated $R = 50$ times with distinct random seeds, yielding a distribution of performance metrics over all Monte Carlo draws from which we report means and standard deviations.

4.1.2 Data Generating Processes

The data are simulated from two DGPs by independently including sparsity in the outcome variables. This yields the two DGPs listed in Table 1.

Table 1: Two simulation scenarios.

No zero inflation	Excess zero inflation
DGP A	DGP B

All DGPs are generated by the same covariate and treatment processes, and they share the same standard outcome distribution. We start by explaining this shared structure, and then describe how the outcome sparsity is added to DGP B. Finally, we will also elaborate on how we construct a contaminated version of DGP B used to stress test a crucial assumption.

Shared structure. For every observation, the treatment assignment is determined by a confounded softmax model. The logit for treatment level $d_k \in \mathcal{D}$ is given by

$$L_{ijt,k} = X_{ijt}^\top \beta_k + c_k, \quad (9)$$

where β_k is drawn uniformly in each replication, and c_k is chosen such that higher discount levels occur less frequently in the treatment assignment. The resulting probability of receiving treatment d_k is calculated as follows: $\pi(d_k \mid X_{ijt}) = \exp(L_{ijt,k}) / \sum_l \exp(L_{ijt,l})$ and D_{ijt} is then a single multinomial draw from $\pi(\cdot \mid X_{ijt})$. Regarding generating the outcome variable, which represents sales, we compute the log value of sales for each treatment level $d_k \in \mathcal{D}$ as follows:

$$\log \mu_{\text{qty}}(d_k, X_{ijt}) = 1.5 + X_{ijt}^\top \delta_{\text{qty}} + 0.2 d_k,$$

where $\delta_{\text{qty}} \in \mathbb{R}^p$ is drawn from a uniform distribution in each replication. The outcomes are then generated from a Negative Binomial distribution with $n = 1$ and $p = (1 + \mu_{\text{qty}}(d_k, X_{ijt}))^{-1}$. This distribution produces zeros with probability p . Consequently, zeros are generated in both DGPs without an explicit hurdle.

Zero-inflation. When simulating DGP B, which incorporates excess zero-inflation in the outcome, we place an additional Bernoulli hurdle in front of the original outcome model described above:

$$Y_{ijt}(d_k) = B_{ijt}(d_k) \cdot Q_{ijt}(d_k), \quad B_{ijt}(d_k) \sim \text{Bernoulli}\left(\sigma(-0.5 + X_{ijt}^\top \gamma + 2.0 d_k)\right),$$

where $Q_{ijt}(d_k)$ is drawn from the same Negative Binomial distribution described in the shared structure, $\sigma(\cdot)$ refers to the logistic function and γ is drawn uniformly in each replication. The hurdle creates a larger point mass on zero representing the zero-inflation often present in a retail context. Note, in DGP A we simply set $B_{ijt}(d_k) = 1$ to avoid this excess zero-inflation.

Spatial interference. Lastly, we introduce a third DGP, which extends DGP B by incorporating spatial interference across units. To this extent, we draw $C = 5$ competitor units $\mathcal{S}_{ij} = \{q_1, \dots, q_C\} \subset \mathcal{I}_j \setminus \{i\}$ without replacement for each unit i in cluster j , and compute an interference signal as follows:

$$\phi_{ijt} = \sum_{r=1}^C \left(c_{\text{disc},r} \cdot D_{q_r,jt} + c_{\text{price},r} \cdot \log \left(\frac{P_{q_r,j}(1 - D_{q_r,jt})}{P_{ij}} \right) \right),$$

where $D_{q_r,jt}$ is the discount applied to peer q_r at time t , and $P_{q_r,j}(1 - D_{q_r,jt})$ is its net price. The coefficients determining the interference strength are drawn independently for each competitor r in each replication, and are chosen such that they resemble economic theory.

Then, this signal is added to the equation for the log-mean of sales:

$$\log \mu_{\text{qty}}(d_k, X_{ijt}, \phi_{ijt}) = 1.5 + X_{ijt}^\top \delta_{\text{qty}} + 0.2 d_k + \phi_{ijt}.$$

This signal shifts the amount of sales based on the prices and discounts of competitors and therefore resembles spatial interference. In DGPs A and B, ϕ_{ijt} is set to 0 to omit these effects. For reference, Appendix C.1 demonstrates figures of the outcome and treatment distribution in DGP A and B.

4.1.3 Estimators

All models estimated on the DGPs from Section 4.1.2 implement the framework in Figure 1. They are characterised by three binary design decisions, resulting in $2^3 = 8$ estimators. These decisions are shown in Table 2.

Table 2: Estimator configurations.

Decision	Option 1	Option 2
First stage architecture	X-learner	HyX-learner
First stage objective	MSE	Tweedie deviance
Regression adjustment	Acharki et al. (2023)	CRC

4.1.4 Evaluation

Since the true CATEs $\tau(d_k, X_{ijt})$ are known by construction in the simulation context, we evaluate all estimators using two scalar metrics computed on the test set. Let the prediction errors of all CATEs on test observations be defined as:

$$e_{ijt,d_k} = \hat{\tau}(d_k, X_{ijt}) - \tau(d_k, X_{ijt}).$$

With these values at all treatment levels we calculate our two evaluation metrics: The Root Mean Squared Error (RMSE) and the bias, given by

$$\text{RMSE} : \sqrt{\frac{1}{\tilde{N}K} \sum_{k=1}^K \sum_{(i,j,t) \in \tilde{\mathcal{U}}} e_{ijt,d_k}^2}, \quad \text{Bias} : \frac{1}{\tilde{N}K} \sum_{k \in \mathcal{U}_k} \sum_{(i,j,t) \in \tilde{\mathcal{U}}} e_{ijt,d_k},$$

where $\tilde{\mathcal{U}}$ denotes the set of test observations, and $\tilde{N} = |\tilde{\mathcal{U}}|$. The RMSE is the primary metric as this provides the best information on the trade-off between bias and variance. The bias shows us whether we systematically over- or underestimate the CATEs. Both metrics are computed for each of the $R = 50$ Monte Carlo replications and summarised by their mean and standard deviation across replications.

4.2 Real-Life Application

This section describes the application of the proposed methods to a real-life data example. This allows us to see how well our models perform in practice, rather than solely in a controlled simulation environment. First, Section 4.2.1 introduces the data we use, and we outline the evaluation approach in Section 4.2.2. This is different from the simulation evaluation since counterfactuals are not known in a real-world context.

4.2.1 Data

We validate the proposed methodology on the Dominick’s Finer Foods cookies scanner dataset, drawn from the Dominick’s Finer Foods Database maintained by the James M. Kilts Center for Marketing at the University of Chicago Booth School of Business¹. The database contains weekly store-level scanner data from Dominick’s Finer Foods, a supermarket chain operating across the Chicago metropolitan area, covering approximately 400 weeks between 1989 and 1994 across a wide range of product categories. As the size of the dataset is large and computing resources are limited, we focus on only the product category labelled as cookies.

Treatment Encoding. Unlike the simulation, the real-life treatment variable is not ordinal but categorical. The dataset records the type of promotion applied, but not the corresponding depth of the price reduction. Treatment is encoded as a discrete variable $D_{ijt} \in \{0, 1, 2, 3\}$:

0 : control, 1 : bonus buy, 2 : price reduction, 3 : coupon.

The CATE $\hat{\tau}(d_k, X)$ measures the incremental expected sales effect of discount mechanism d_k relative to no discount. Despite the fact that our proposed methods were originally designed for an ordinal discount depth, they only require treatment levels to be discrete. In the pooled model, treatment is encoded using one-hot encoding. Therefore, this dataset allows for a valid assessment of our methods’ performance in a real-life setting.

¹Available at www.chicagobooth.edu/research/kilts/research-data/dominicks

Covariates. The covariate vector X_{ijt} contains the eleven variables described below. Note, that the outcome of interest Y_{ijt} is the sales of unit i in store j during week t .

<code>log(unit_price_{ijt})</code> ,	log of the price, capturing price sensitivity of sales.
<code>size_oz_{ij}</code> ,	product size in fluid ounces, a product attribute which is constant over time.
<code>log_case_{ij}</code> ,	log case pack quantity, reflecting the number of items per case as the supermarket receives them from the manufacturer.
<code>week_of_year_t</code> , $\in \{0, \dots, 51\}$,	the position within the annual cycle, capturing possible weekly seasonality in cookie demand.
<code>year_t</code> $\in \{0, 1, 2, \dots\}$,	a trend variable, capturing demand changes.
Store demographics,	median household income (<code>income</code>), average household size (<code>hsizeavg</code>), population density (<code>density</code>), urban fraction (<code>urban</code>), fraction of low-income households (<code>priclow</code>), and fraction of high-income households (<code>prichigh</code>).

Sampling. The raw transaction file contains more than 13 million valid observations after quality filtering. As Random Access Memory (RAM) is limited, we need to accounting for computational limits during model fitting. Hence, we choose to draw a random sample of 15% of all valid observations (approximately 2 million records). Sampling is performed with a fixed random seed to ensure reproducibility.

Training Split. Data are split on the week variable. All observations from the earliest 80% of the weeks form the training set. The remaining 20% of the weeks form the test set. This is similar to the split used in the simulation study for avoiding look-ahead bias. For reference, Appendix C.2 demonstrates figures of the outcome and treatment distribution in the dataset.

4.2.2 Evaluation

Since the true CATEs are unknown in real-world data, determining the RMSE is impossible. We instead use the Grouped Average Treatment Effects (GATES) test of Chernozhukov et al. (2018b) to assess whether the estimated CATE rankings are consistent with the heterogeneity of the realised treatment effect. The test examines whether the quintile with the highest predicted response to treatment also exhibits a larger observed average treatment effect than those in the lowest quintile. This is a necessary condition for valid CATE estimation. For each treatment arm $d_k \in \{1, 2, 3\}$ and each of the eight estimators, we apply the following procedure to the test set.

Score and rank. Compute the predicted CATE $\hat{\tau}(d_k, X_i)$ for every test observation i . Assign each observation to a quintile group $G_i \in \{1, \dots, 5\}$ based on the rank of $\hat{\tau}(d_k, X_i)$ among all test observations. They are ranked so that $G_i = 1$ identifies the bottom 20% of predicted treatment responsiveness and $G_i = 5$ the top 20%. All observations are assigned to one of the quintiles, so also the observations that are not assigned to the treatment of interest.

Filter. Now we restrict the analysis to observations receiving either the control or the target treatment, creating the sample $\mathcal{T}_k = \{i : D_{ijt} \in \{0, d_k\}\}$ that we will use for analysis.

GATES regression. Estimate the following linear model on $\mathcal{T}_k$ with ordinary least squares:

$$Y_{ijt} = \sum_{g=1}^5 \alpha_g \mathbf{1}[G_i = g] + \sum_{g=1}^5 \gamma_g \mathbf{1}[G_i = g] (W_i - \bar{p}) + \varepsilon_i,$$

where $W_i = \mathbf{1}[D_{ijt} = d_k]$ is the binary treatment indicator, and $\bar{p} = |\{i \in \mathcal{T}_k : W_i = 1\}|/|\mathcal{T}_k|$ is the empirical treatment rate in $\mathcal{T}_k$. The group intercepts α_g absorb mean outcomes per group, and the demeaning $W_i - \bar{p}$ ensures orthogonality between the intercept and slope terms. The coefficient γ_g then identifies the average treatment effect within quintile g .

Hypothesis test. Test whether quintile number five has a strictly higher average treatment effect than quintile number one:

$$H_0 : \gamma_5 \leq \gamma_1 \quad \text{vs.} \quad H_1 : \gamma_5 > \gamma_1,$$

using a one-sided t -test with $|\mathcal{T}_k| - 2G$ degrees of freedom. The test statistic is $t = (\hat{\gamma}_5 - \hat{\gamma}_1)/\widehat{\text{se}}$, where $\widehat{\text{se}}$ is the standard error of the difference $\hat{\gamma}_5 - \hat{\gamma}_1$. This standard error is obtained as $\hat{\sigma}_{55} + \hat{\sigma}_{11} - 2\hat{\sigma}_{51}$. We report the point estimate $\hat{\gamma}_5 - \hat{\gamma}_1$, its standard error, the t -statistic, and the one-sided p -value in Section 5.

Rejection of H_0 at a 5% significance level provides evidence that the CATE estimator successfully ranks units by their responsiveness to the treatment. Failure to reject indicates either genuinely flat treatment effect heterogeneity or a failure of the CATE estimation model to rank units correctly. The procedure is applied independently to each of the three treatment arms ($d_k \in \{1, 2, 3\}$) and each of the eight estimators, yielding a 3×8 matrix of the GATES test statistics.

5 Numerical Results

This section provides analyses on the results of the simulation study and the real-life application to the Dominick’s Finer Foods cookies data. All experiments are executed on a Macbook Pro (2024) with an M4 processor, 16GB of RAM, running MacOS Sequoia 15.7.3. All code, including documentation and data can be found in a public repository². The remainder of this section is structured as follows. Section 5.1 evaluates all eight estimation methods against known ground-truth CATEs across $R = 50$ Monte Carlo replications on DGP A and DGP B. Section 5.2 evaluates the estimators on the Dominick’s dataset using the GATES framework.

5.1 Simulation Results

To demonstrate the numerical results and findings of our simulation study, we adhere to the following structure. First, we evaluate the overall prediction accuracy of the CATEs by comparing the RMSE values and the biases in Section 5.1.1. Then, Section 5.1.2 assesses the added value of implementing the HyX-learner framework, by examining the performance per treatment level. Lastly, Section 5.1.3 conducts a stress test against the SUTVA assumption in (2), using data contaminated with spatial interference.

5.1.1 Overall CATE Accuracy

We performed $R = 50$ Monte Carlo replications, and report the mean and standard deviation of the RMSEs of the CATE estimates in Table 3. The results will be discussed in order of their contribution to accuracy. Subsequently, we discuss the distributions of biases across the Monte Carlo replications demonstrated in Figure 2. These plots provide additional support for the RMSE patterns in Table 3.

Table 3: RMSE values for every model and DGP combination.

Model	RA Formula	Objective*	DGP A	DGP B
X-learner	Acharki et al. (2023)	MSE	3.809 (0.79)	3.658 (0.78)
		Tweedie	4.285 (0.98)	4.233 (0.91)
	CRC	MSE	3.790 (0.86)	3.572 (0.79)
		Tweedie	3.178 (0.68)	2.939 (0.63)
HyX-learner	Acharki et al. (2023)	MSE	1.970 (0.51)	2.546 (0.66)
		Tweedie	2.367 (0.59)	3.068 (0.75)
	CRC	MSE	1.507 (0.35)	2.046 (0.55)
		Tweedie	1.449 (0.31)	2.105 (0.50)

* Applied to the first stage outcome models.

²Available at: <https://github.com/TumGommans/causal-forecasting>. Instructions on how to replicate all results are written in ./README.md

Main Findings. Four concrete conclusions can be derived from the results in Table 3. First, the HyX-learner consistently outperforms the X-learner across all possible settings. Second, the proposed CRC provides consistent improvements in RMSE compared to the regression adjustment formula proposed by Acharki et al. (2023). Third, the choice of objective function is highly dependent on the regression adjustment formula. When applying the Tweedie loss formula instead of MSE, predictive performance turns out worse when this is combined with the Acharki et al. (2023) formula. However, the Tweedie loss consistently decreases RMSE when paired with our proposed CRC, relative to using the MSE objective. Finally, we conclude that optimality for both DGPs is reached with the HyX-learner in combination with the CRC. In particular, the Tweedie loss performs better in DGP A ($\text{RMSE } 1.449 < 1.507$), and under DGP B the MSE performs slightly better ($\text{RMSE } 2.046 < 2.105$). Figure 2 visualises the patterns by showing systematic bias and RMSE together in the different scenarios.

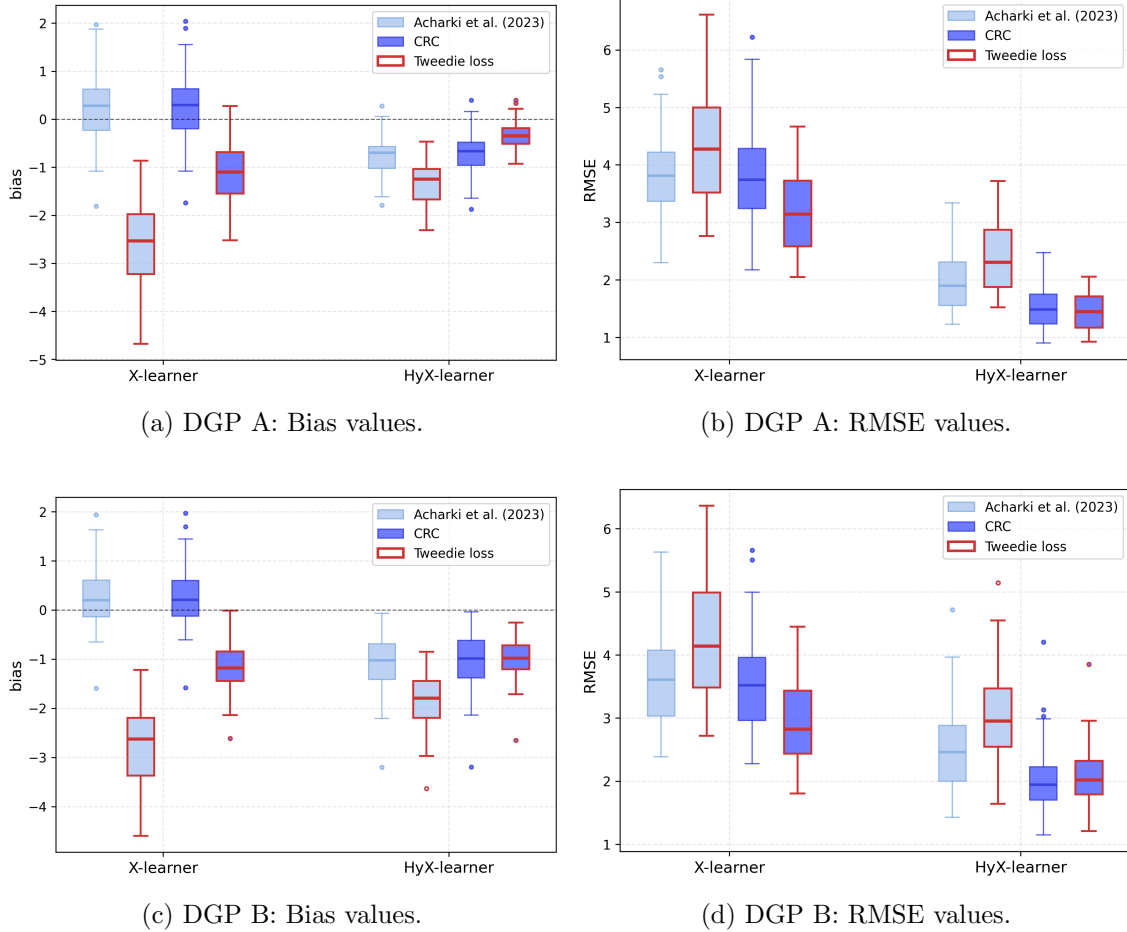


Figure 2: Performance metrics across $R = 50$ replications.

Model. The advantage of the HyX-learner over the X-learner comes from its variance reduction mechanism. The HyX-learner is more robust in sparse groups because it pools data across all treatment levels. This mechanism significantly reduces the noisy estimation errors present in the standard X-learner, leading to the substantial drop in RMSE observed under DGP A. The effect will be illustrated further in Section 5.1.2. While the HyX-learner continues to outperform the X-learner under DGP B, the transition from DGP A reveals an interesting divergence. The HyX-learner’s performance declines when moving to the zero-inflated distribution, whereas the X-learner shows slight improvements. This shift likely follows from the complex relationship between treatment assignment and the zero-inflated outcome. As stated in Section 4.1.2, the probability of outcomes being zero depends on the treatment assignment. Because the S-learner component of the HyX-learner pools data globally, it may struggle to isolate treatment-specific signals. The X-learner models each stratum separately, and therefore does not have this problem. The high frequency of zeros in the outcome seems to actually stabilize the X-learner’s predictions. This reduction in predictive noise allows the model to be more consistent, ultimately leading to a lower RMSE. Further support for this result is provided by the moderately equal bias distributions in Figures 2a and 2c.

RA Formula & Objective. The results in Table 3 and the bias distributions in Figure 2 demonstrate a relationship between the regression adjustment formula and the objective function. When using the regression adjustment of Acharki et al. (2023), the Tweedie objective uniformly worsens the RMSE. For instance, when using the original Generalised X-learner in DGP A, the Tweedie loss increases the X-learner’s RMSE by approximately 12.5% and the HyX-learner’s RMSE by more than 20%. The bias distributions in Figures 2a and 2c reveal why these scenarios are less important for the final framework. The Tweedie deviance function reduces the weights of large observations during training, resulting in under-predictions for stage 1 models, and therefore large positive residuals. In the original Generalised X-learner framework these positive residuals are subtracted from the imputed treatment effect, pushing pseudo-outcomes downward for the units with the highest CATEs. This introduces the systematic negative bias that is visible, for example, in plot 2a for the X-learner with the Tweedie loss function and the original formula proposed by Acharki et al. (2023). Our proposed CRC mathematically inverts this error, neutralising this systematic bias. Furthermore, the CRC shows its full potential within the environment with low variance created by the HyX-learner. Because the HyX-learner mitigates variance through pooling, bias becomes the dominant source of remaining error. By directly addressing this bias, using the CRC in the HyX-learner with Tweedie loss achieves the optimal bias-variance trade-off.

5.1.2 Heterogeneity Across Discount Depths

Although the RMSE values in Table 3 summarize overall accuracy, examining performance by treatment level reveals where the HyX-learner provides the greatest improvement. Figure 3 shows the RMSE across discount levels for both learners. The results use the best performing configuration from Table 3 under DGP B. The figure illustrates how the pooling mechanism stabilizes estimation when treatment groups become sparse.

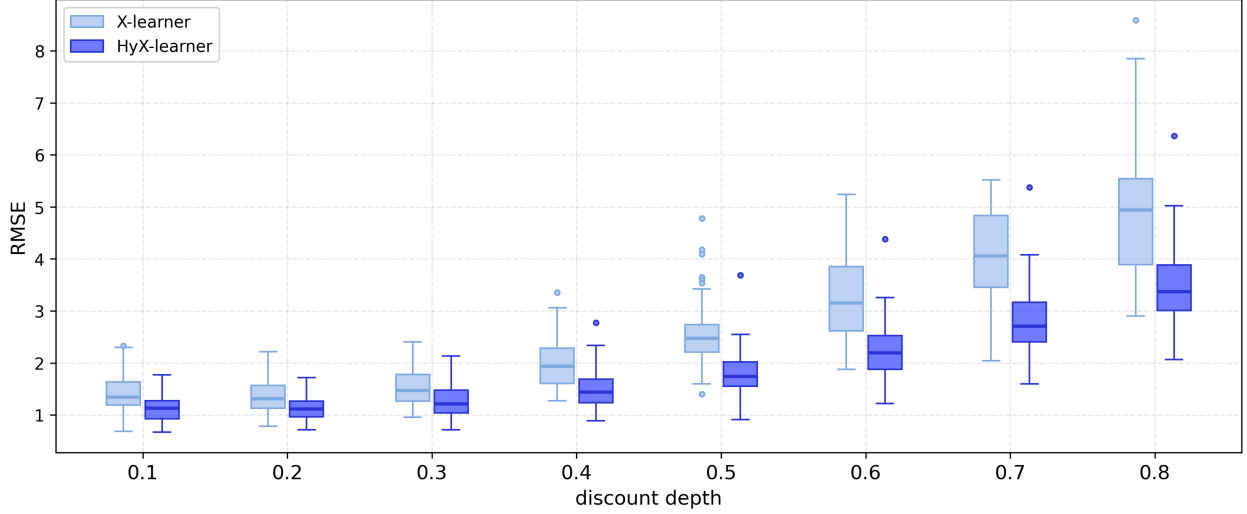


Figure 3: Per-treatment RMSEs for the estimators with CRC and Tweedie loss.

As illustrated above, the advantage of the HyX-learner increases as the discount depth becomes larger. For smaller discount levels, the two learners perform similarly. However, as discount levels increase, the purely stratified X-learner shows a clear rise in both its median RMSE and its interquartile range. In contrast, the HyX-learner’s performance worsens less strongly for deeper discounts. This difference follows directly from the structural sparsity created by the treatment assignment design in (9). The decreasing intercepts concentrate most observations at low discount levels and leave the deep discount arms with very few observations. For these sparse groups, the combination weight of the HyX-learner adjusts automatically ($\alpha_k \rightarrow 0$ as $n_k \rightarrow 0$). This shifts the estimation from the highly variable stratified model to the more stable pooled model that uses the full sample. As a result, the model behaves in line with the James Stein shrinkage principle discussed in Section 3.2. Ultimately, the much narrower interquartile range of the HyX-learner at deep discount levels supports the main idea of the framework, that is, the pooling mechanism stabilizes estimation when treatment groups become sparse.

5.1.3 Robustness to Spatial Interference

As discussed in Section 2, standard causal inference relies heavily on the SUTVA assumption. This assumption states that the treatment assigned to one unit does not affect the outcomes of other units. In retail environments, however, this assumption is often violated because of spatial

substitution and competitor pricing dynamics. To examine how robust our framework is to such violations, Table 4 reports a stress test in which DGP B is extended with a spatially correlated interference term, ϕ_{ijt} .

Table 4: Bias and RMSE values for the interference stress test.

Model	RA Formula	Objective *	No interference		Interference	
			Bias	RMSE	Bias	RMSE
X-learner	Acharki et al. (2023)	MSE	0.262 (0.61)	3.658 (0.78)	-1.017 (0.70)	6.173 (3.84)
		Tweedie	-2.777 (0.72)	4.233 (0.91)	-3.148 (0.66)	5.594 (1.72)
	CRC	MSE	0.271 (0.61)	3.572 (0.79)	-1.020 (0.65)	5.462 (3.55)
		Tweedie	-1.173 (0.50)	2.939 (0.63)	-1.864 (0.53)	3.893 (1.10)
HyX-learner	Acharki et al. (2023)	MSE	-1.108 (0.58)	2.546 (0.66)	-1.207 (0.67)	5.502 (3.47)
		Tweedie	-1.877 (0.57)	3.068 (0.75)	-3.044 (0.64)	5.267 (1.59)
	CRC	MSE	-1.062 (0.57)	2.046 (0.55)	-1.191 (0.62)	5.008 (3.42)
		Tweedie	-1.001 (0.44)	2.105 (0.50)	-1.760 (0.51)	3.847 (1.22)

* Applied to the first stage outcome models.

The main result from Table 4 is that introducing spatial interference strongly reduces the performance of all estimator configurations. Unlike simple cluster-level shifts, this localized interference acts as a strong and unobserved structural signal. For the X-learner under the MSE objective and CRC, RMSE increases sharply from 3.572 to 5.462. The HyX-learner shows a similar pattern, with RMSE rising from 2.046 to 5.008. In addition, all models experience a clear deterioration in mean bias, which becomes strongly negative.

This drop in performance reflects the structure of the spatial signal ϕ_{ijt} . The interference is generated through random interactions with a subset of competitor products. Because of this, the resulting variation cannot be captured by the unit-level or cluster-level covariates contained in X_{ijt} . In other words, XGBoost is unable to learn or absorb this sales shock. As a result, the interference is either treated as extra noise, which increases the variance of the stage 1 models, or it is wrongly attributed to the treatment assignment. This produces systematic negative bias. We do observe that the Tweedie objective handles this injected noise somewhat better than the MSE objective. For example, the HyX-learner with Tweedie yields an RMSE of 3.847 rather than 5.008. However, the overall error levels remain high.

Overall, the HyX-learner still retains a small predictive advantage over the standard X-learner when interference is present. For example, under CRC the Tweedie RMSE is 3.847 compared to 3.893. However, the difference between the two models becomes much smaller. The pooling mechanism helps stabilize estimation when treatment groups are sparse, but it cannot overcome severe violations of the SUTVA assumption. This stress test therefore underscores an important limitation, regardless of the meta-learner and model configuration used. Although not addressed in our main research, we drafted several ways on how to capture interference signals in the covariate space in Appendix D.

5.2 Real-Life Application

The Dominick’s Finer Foods application presents a different evaluation challenge. Each unit is observed under only one treatment level. This means the potential outcomes $Y_{ijt}(d_k)$ for $d_k \neq D_{ijt}$ cannot be observed. Because of this, simulation-based RMSE and bias metrics cannot be computed. It would therefore be incorrect to interpret the results below as evidence of predictive accuracy. Instead, we apply the GATES framework described in Section 4.2.2. This method tests whether a model’s CATE rankings are economically meaningful. It does so by comparing the average treatment effects between the top and bottom quintiles of predicted CATE values. A significantly positive statistic $\hat{\gamma}_5 - \hat{\gamma}_1 > 0$ indicates that units predicted to be most responsive indeed show larger treatment effects. This provides a measure of ranking validity rather than absolute predictive accuracy. Table 5 reports these statistics for all three treatments.

Table 5: Extreme groups test statistics: $(\hat{\gamma}_5 - \hat{\gamma}_1)$.

Model	RA Formula	Objective*	Bonus Buy	Price Reduction	Coupon
X-learner	Acharki et al. (2023)	MSE	-0.20 (0.55)	43.95*** (3.71)	138.95*** (9.86)
		Tweedie	-2.13 (0.51)	47.60*** (2.34)	112.33*** (14.63)
	CRC	MSE	2.00*** (0.50)	39.19*** (3.76)	138.63*** (9.81)
		Tweedie	5.36*** (0.43)	48.28*** (2.18)	125.49*** (8.47)
HyX-learner	Acharki et al. (2023)	MSE	0.12 (0.54)	43.46*** (3.84)	129.36*** (13.92)
		Tweedie	-1.65 (0.50)	46.52*** (2.25)	124.38*** (8.44)
	CRC	MSE	0.87** (0.48)	39.88*** (3.81)	133.04*** (9.57)
		Tweedie	4.90*** (0.41)	47.42*** (2.14)	124.35*** (8.35)

Heteroskedasticity-robust standard errors in parentheses.

*** $p < 0.01$, ** $p < 0.05$.

* Applied to the first stage outcome models.

The Bonus Buy treatment arm provides the most informative test case, as it is the only arm where the choice of regression adjustment formula changes the inference. The uncorrected framework of Acharki et al. (2023) fails to detect meaningful heterogeneity in this setting. The MSE configurations produce statistically insignificant test statistics (-0.20 and 0.12), while the Tweedie variants generate significantly negative statistics (-2.13 and -1.65). A negative test statistic indicates an inverted ranking, meaning that units predicted to be the least responsive actually show the largest realized treatment effects. This inversion arises from a combination of errors rather than simple underprediction. The Tweedie objective naturally down-weights high-variance, high-sales observations during stage-1 training. As a result, the model systematically underpredicts outcomes for inherently high-volume units. These units therefore generate the largest positive residuals. Because the Acharki et al. (2023) pseudo-outcome formula subtracts these residuals, it imposes the strongest downward penalty on the units with the highest baseline sales and true responsiveness. This heterogeneity penalty distorts the relative CATE predictions, pushing units with the largest treatment effects into the lowest predicted quintiles and reversing the GATES test statistic.

Implementing CRC removes this distortion and restores the validity of the rankings. All four corrected variants produce statistically significant and positive test statistics for the Bonus Buy treatment. Notably, the Tweedie objective performs substantially better than MSE when combined with CRC. It produces much larger test statistics (5.36 for the X-learner and 4.90 for the HyX-learner). This improvement highlights the advantage of the Tweedie distribution when modelling the heavily zero-inflated outcomes typical for this promotion type.

In contrast, the Price Reduction and Coupon treatments show sufficiently strong underlying heterogeneity that even misspecified estimators are able to detect it. All eight estimator configurations produce highly significant positive GATES statistics ($p < 0.01$) for these treatments. The quintile differences range from about 39 to 48 units for Price Reduction and from 112 to 139 units for Coupon. The large effects in the Coupon treatment likely reflect strong differences in responsiveness across store demographics. Nevertheless, the choice of model configuration still influences ranking accuracy. For the Price Reduction treatment, the CRC Tweedie configurations produce the largest quintile separations (48.28 for the X-learner and 47.42 for the HyX-learner), outperforming their MSE counterparts. For the Coupon arm, however, the MSE configurations yield slightly larger test statistics. This suggests that the MSE objective may better capture the outcome structure of coupon redemptions. Despite these differences, the uncorrected Tweedie specifications based on the formulation of Acharki et al. (2023) consistently perform worse than the CRC variants across all treatment arms. This confirms the continued presence of the negative bias amplification.

The contrast between the Bonus Buy treatment and the Price Reduction treatment highlights the main empirical contribution of the proposed methodology. When treatment effect heterogeneity is strong, standard methods are generally able to detect it. However, more challenging retail settings require additional methodological care. These settings are often characterized by smaller effects, low signal-to-noise ratios, and zero-inflated outcomes. The Dominick’s Finer Foods application shows that CRC and the Tweedie objective are important for maintaining reliable causal inference in these noisy retail environments.

6 Conclusion

Retailers increasingly use data-driven methods to optimise discount policies, however, estimating the causal effect of a discount on sales remains challenging. This paper studied heterogeneous treatment effect estimation with multi-level treatment under the challenges found in retail data: zero-inflated outcomes, skewed sales distributions and under-represented treatment groups. To address these challenges, we made three contributions: First, we introduced the HyX-learner which builds upon the Generalised X-learner by combining stratified models with a pooled model, reducing estimation variance in sparse treatment groups. Second, we incorporated the Tweedie objective to handle the zero-inflation and skewness of retail data. Third, we proposed CRC as a correction of the regression adjustment formula of the Generalised X-learner.

Our simulation study showed that the HyX-learner consistently outperformed the X-learner across both standard and zero-inflated data generating processes. Moreover, we found that this difference was the largest for sparse treatment groups. In addition, we found that the Tweedie objective improved performance when paired with CRC, while pairing it with the Acharki et al. (2023) adjustment yielded worse results compared to MSE. This is due to the compounding of negative bias in both the Tweedie loss function and the Acharki et al. (2023) regression adjustment. Furthermore, we found that CRC consistently outperformed the adjustment found in Acharki et al. (2023). Using the GATES framework, the real-life application showed that for CRC both the HyX-learner and the X-learner correctly rank observations based on predicted CATE, where the Acharki et al. (2023) adjustment produced inverted rankings across all configurations for the Bonus Buy treatment.

Several interesting avenues for further research arise from this paper. Our stress test revealed that when the SUTVA assumption is violated the performance of all models degrades, although the HyX-learner with Tweedie loss and CRC still yielded the best results. Therefore, a natural extension would be incorporating spatial interference. In addition, modelling intertemporal interference through dynamic treatment effects would better account for the stockpiling effects that are common in retail. Lastly, extending the framework to continuous discounts would make the model applicable in settings without discrete treatment levels.

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A Hyperparameters

A.1 Combination Weights

We select δ by 5-fold cross-validation on the training data. For each fold $f \in \{1, \dots, 5\}$, the training data are split into a training set $\mathcal{F}_f$ and a validation set $\mathcal{V}_f$. On $\mathcal{F}_f$ we fit one stratified model $\hat{\mu}_{d_k}^{(f)}$ per treatment arm and one pooled S-learner $\hat{\mu}^{(S,f)}$ jointly. Both sets of models are fitted once per fold, independently of δ . Candidate values are then evaluated numerically. The selection criterion is the MSE over all non-control treatment levels and all folds.

Candidate grid. The grid spans several orders of magnitude around the baseline value $\delta = 1$, which corresponds to the hyperparameter-free formula for α_k :

$$\mathcal{G} = \{1.0, 2.0, 5.0, 10, 20, 50, 100, 200, 500, 1000\}.$$

The lower boundary is the natural benchmark at which the weight formula requires no additional calibration. The upper boundary ($\delta = 1000$) effectively collapses the combination to the pooled model for any stratum whose size is not orders of magnitude above the average, and is included to detect whether the data strongly favour pure pooling.

Protocol. In the simulation study, $\hat{\delta}$ is selected once per DGP and estimator configuration on the training partition of the dataset generated with the base random seed $r = 42$. The resulting value is then held fixed across all $R = 50$ Monte Carlo replications.

Selected values. Table 6 reports the $\hat{\delta}$ values obtained for the simulation study. The CV consistently selects large values, implying strong shrinkage toward the pooled model across all but the most common treatment arms.

Table 6: Cross-validation selected $\hat{\delta}$ values per DGP and estimator configuration.

Model	RA Formula	Objective	DGP A	DGP B
HyX-learner	Acharki et al. (2023)	MSE	100	500
		Tweedie	200	1000
	Residual correction	MSE	100	500
		Tweedie	200	1000

A.2 Base Learners

All regression tasks within the X-learner and HyX-learner are implemented as XGBoost regressors. The hyperparameters listed in Table 7 were set as conservative defaults prior to all experiments, and are held constant across all model configurations, DGPs, and Monte Carlo replications within each setting. No grid search was performed as the focus of the paper is on the causal meta-learning framework rather than optimising the predictive performance of base learners in isolation. Motivation for the values in Table 7 is given below.

Table 7: XGBoost hyperparameters for the simulation study and the real-life application.

Parameter	Description	Simulation	Real-life
<code>n_estimators</code>	Number of boosting rounds	100	200
<code>max_depth</code>	Maximum depth per tree	3	4
<code>learning_rate</code>	Step-size shrinkage (η)	0.05	0.05
<code>min_child_weight</code>	Minimum sum of instance weights in a leaf	10	100
<code>subsample</code>	Fraction of training rows sampled per tree	0.8	0.8
<code>colsample_bytree</code>	Fraction of features sampled per tree	0.8	1.0

Tree depth and minimum child weight. Shallow trees paired with a minimum child weight provide structural regularisation that is especially important for the Stage 1 stratified models, where each arm’s training set is a strict subset of the full data. Without this constraint, models fitted on sparse treatment arms would routinely overfit to their small strata, inflating the variance of the Stage 2 pseudo-outcomes through the error-propagation mechanism. In the real-life application the training set is of larger magnitude and the underlying outcome is more complex. Accordingly, `max_depth` is increased to 4 and `min_child_weight` is set to 100, relatively maintaining a comparable degree of regularisation per leaf.

Learning rate and number estimators. A learning rate of $\eta = 0.05$ combined with 100 estimators in the simulation and 200 in the real-life setting provides a well-regularised boosting schedule without prohibitive runtime. Gradient boosted trees with a small step size and sufficiently many rounds are less sensitive to early stopping and overfit slower than higher learning rates, which is desirable in Stage 1 where the training data per arm can be limited. The larger number of rounds in the real-life setting compensates for the smaller per-step update and ensures the ensemble can fit the more complex outcome structure of the Dominick’s cookies data.

Subsampling. Row and column subsampling rates of 0.8 introduce stochasticity at each boosting step, reducing variance and providing regularisation analogous to dropout in neural networks. In the real-life application, `colsample_bytree` is set to 1.0 because the covariate space contains only eleven features. Sampling a fraction of columns at this dimensionality would discard too large a share of the available signal in each tree and could cause relevant predictors to be systematically excluded.

B Simulation Details

This appendix supplies the distributional parameters of the data-generating processes in Section 4.1.2 that are not stated in the main text. Each Monte Carlo replication uses a distinct integer seed $r \in \{0, 1, \dots, 49\}$. The combination weight $\hat{\delta}$ is cross-validated on the training partition of the replication with base seed $r = 42$, which is also used as the first replication (see Appendix A.1).

Treatment assignment. The coefficients in the confounded softmax model are drawn as follows in each replication, where the log-price covariate is excluded from the treatment logits to prevent its non-zero population mean from distorting the intercept structure:

$$\begin{aligned}\beta_k &\sim \text{U}(-0.5, 0.5)^{19}, \quad k = 0, \dots, 8, \\ (c_0, \dots, c_8) &= (3.000, 2.375, 1.750, 1.125, 0.500, -0.125, -0.750, -1.375, -2.000),\end{aligned}$$

evenly spaced on $[-2, 3]$ so that treatment frequency decreases monotonically with discount depth.

Outcome model. The coefficient vector $\delta_{\text{qty}} \in \mathbb{R}^{20}$ is drawn as follows in each replication, with the Negative Binomial dispersion $\alpha = 1$ fixed throughout:

$$\begin{aligned}\delta_{\text{qty}, 1:19} &\sim \text{U}(0, 0.15)^{19}, \\ \delta_{\text{qty}, 20} &\sim \text{U}(-0.3, -0.1). \quad (\text{log-price coefficient})\end{aligned}$$

Zero-inflation. The coefficient vector $\gamma \in \mathbb{R}^{20}$ is drawn as follows in each replication:

$$\begin{aligned}\gamma_{1:10} &\sim \text{U}(-0.5, 0.5)^{10}, \\ \gamma_{11:20} &= \mathbf{0}.\end{aligned}$$

Spatial interference. The interference coefficients are drawn independently per competitor $r \in \{1, \dots, 5\}$ in each replication. Setting $\xi = 3$, the base draws are scaled as $c_{\text{disc}, r} = \xi \cdot u_{\text{disc}, r}$ and $c_{\text{price}, r} = \xi \cdot u_{\text{price}, r}$, yielding:

$$\begin{aligned}u_{\text{disc}, r} &\sim \text{U}(-1.2, -0.6), \\ u_{\text{price}, r} &\sim \text{U}(0.3, 0.6).\end{aligned}$$

The negative sign on $c_{\text{disc}, r}$ encodes substitution effects; the positive sign on $c_{\text{price}, r}$ encodes a relative-price effect. The scaling $\xi = 3$ is chosen so that the resulting ϕ_{ijt} is of comparable magnitude to the direct treatment effect $0.2 d_k$, constituting a practically severe SUTVA violation.

C Data Characteristics

C.1 Simulated Data

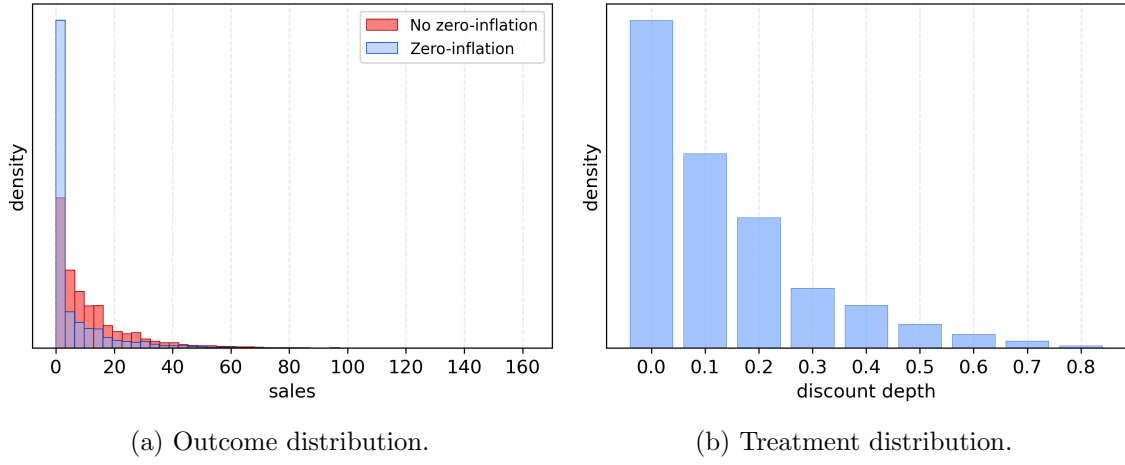


Figure 4: Simulated data plots for 1 Monte Carlo replication.

C.2 Real Data

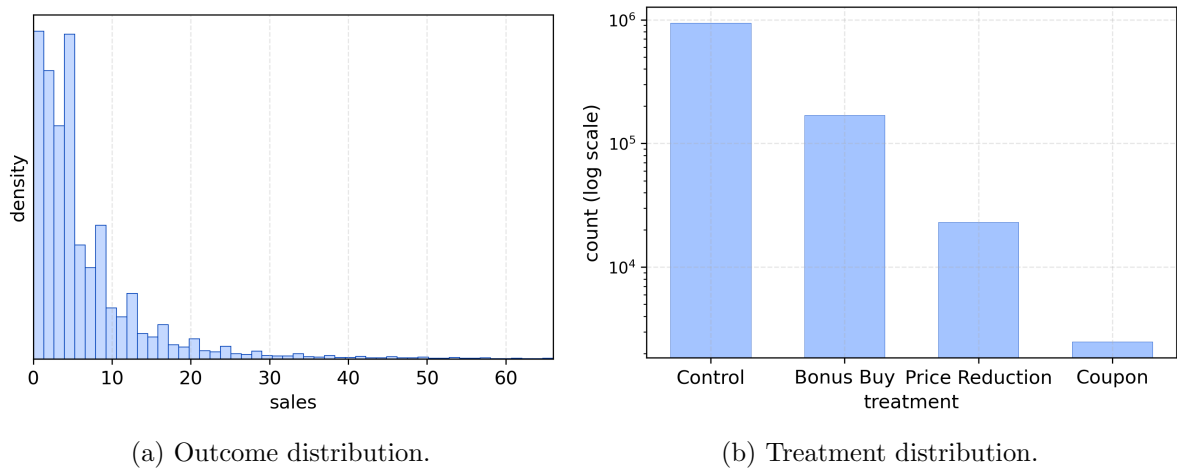


Figure 5: Real data plots.

D Interference

D.1 Spatial Interference

Retail sales exhibit spatial interference as sales arise not only from the need for a specific unit, but also from the need for a product category. For example, customers in need of a jacket typically do not buy more than one, and therefore interference exists between similar units. By extension, discounts of one unit may lead to decreased sales of other units, and ignoring this dependence leads can result in biased estimates of the CATE.

To tackle this potential bias, we augment the conditioning statement in (3), which corresponds to the first stage of the X-learner, with additional information regarding the interference. To this extent, we assume that interference of unit $i \in \mathcal{I}_j$ only depends on units in the same cluster $\mathcal{I}_j$. Given this, we define D_{ijt}^c to be the maximum promotional activity within a cluster and P_{ijt}^c as the minimum price of a unit within a cluster. Mathematically, D_{ijt}^c and P_{ijt}^c are given by,

$$D_{ijt}^c = \max_{q \in \mathcal{I}_j \setminus \{i\}} \{D_{qjt}\}, \quad P_{ijt}^c = \min_{q \in \mathcal{I}_j \setminus \{i\}} \{P_{qjt}(1 - D_{qjt})\}$$

In our interference aware estimators, we include D_{ijt}^c and P_{ijt}^c in the observed characteristics X_{ijt} . This approach relies on two main assumptions. First, we assume that interference only takes place within clusters. With properly defined clusters this is a reasonable assumption in retail, as discounting a jacket is unlikely to deter customers from buying shoes. Second, we assume that the aggregates P_{ijt}^c and D_{ijt}^c are sufficient for estimating the effects of competing discounts. This is an approximation, as in reality customers might respond to the joint distribution of prices rather than the minimum or maximum.

D.2 Temporal Interference

In addition to spatial interference, retail sales could suffer from intertemporal interference due to customer stockpiling, demand displacement and category satisfaction. Past discounts can therefore affect current sales, even when there is no contemporaneous discount. We initially included a mechanism to account for this problem in our SIX-learner. We decided to exclude it in the final framework to maintain the focus on general retail scenarios. The phenomenon of stockpiling is only present in specific retail applications (e.g., supermarkets) among specific SKUs (e.g., soap). Besides that, the amount of intertemporal interference is rather heterogeneous across customers than across SKUs, and our framework focusses on SKU level data.

We did already design a method for tackling intertemporal interference. Rather than solving a dynamic optimisation problem, we introduce a reduced-form approach in which we treat past promotions and past sales as state variables. In this manner, we incorporate intertemporal dependencies while maintaining a static framework.

We summarize past promotional activity D_{ijt}^{lag} and previous sales Y_{ijt}^{lag} as a weighted average of previous discounts and sales, respectively:

$$D_{ijt}^{\text{lag}} = \sum_{s=1}^L \lambda_s D_{ij,t-s}, \quad Y_{ijt}^{\text{lag}} = \sum_{s=1}^L \omega_s Y_{ij,t-s},$$

where L denotes the backward-looking horizon for both discounts and sales. Moreover, the weights $\lambda_s, \omega_s \geq 0$, satisfy $\sum_{s=1}^L \lambda_s = 1, \sum_{s=1}^L \omega_s = 1$, and decay exponentially:

$$\lambda_s = \frac{\exp(-\gamma_\lambda s)}{\sum_{l=1}^L \exp(-\gamma_\lambda l)}, \quad \omega_s = \frac{\exp(-\gamma_\omega s)}{\sum_{l=1}^L \exp(-\gamma_\omega l)}. \quad (10)$$

Through (10) we model that recent promotions are more influential than old promotions. We include D_{ijt}^{lag} and Y_{ijt}^{lag} in the set of observed characteristics X_{ijt} when estimating the CATE. By conditioning on past discounts and past sales we approximate the demand state due to previous discounts without explicitly modelling a dynamic programming problem. Note that this does not solve the whole problem of intertemporal interference. The decision rule is myopic, as it solely takes into account past discounts and sales, and not its own effect on future sales. This proposed method could be useful in specific retail scenarios, and the role of temporal interference is a promising area for additional research.

E Artificial Intelligence Statement

Generative AI tools have solely been used for minor polishing of academic writing as well as code documentation polishing. To this extent, we made use of `gemini-3.1-pro` and `claude-sonnet-4.6`.